

Problem
Statement 06:
 Accessibility &
 Inclusive

Technology

Shram Saathi: Zero-App Voice AI Middleware

Bridging accessibility and language barriers via WhatsApp, IVR, and CSC using BHASHINI & 3-Layer Authentication.

PROBLEM DATA: Crores of illiterate workers are unable to use these government portals due to text-only interfaces, language barriers, & high biometric failure rates (Jharkhand 49%, Rajasthan 37%).

Building: "Shram Saathi"

Using

Zero Setup: No app downloads required.

Omnichannel: Accessible via WhatsApp, IVR & CSC.

Direct Govt API: GIGW 3.0 & DPDP compliant linking.

Secure Processing:
Auto-masked data processing.

Proven Tech: Powered by Govt's successful BHASHINI framework.

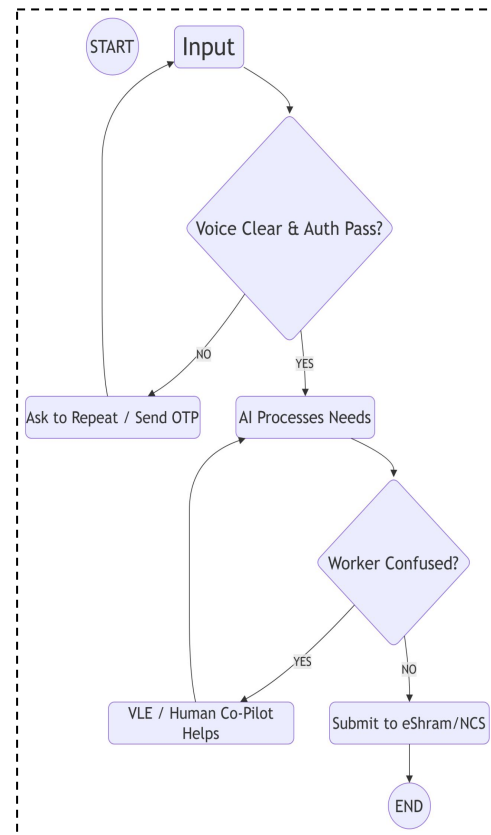
Uses

Native Language Access: eShram, PF & ESI everything by just speaking.

Biometric Fallback: Voice, OTP gets the job done when fingerprints fail.

Zero Friction: As simple as WhatsApp strictly no downloads required.

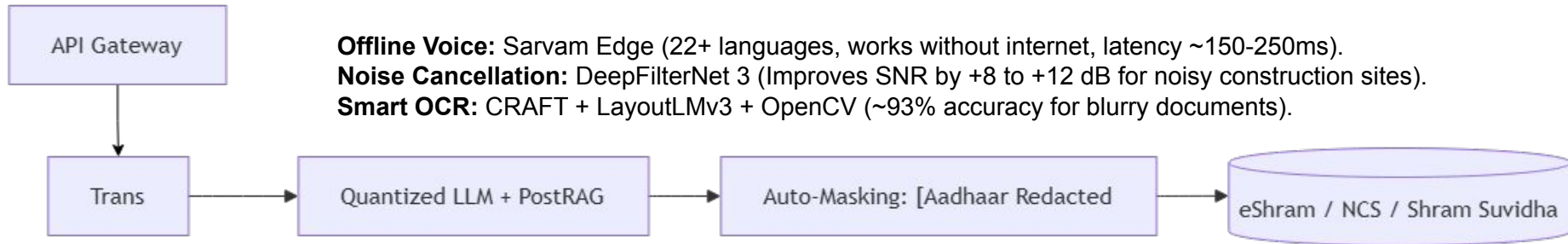
Human-in-the-Loop: No worker will ever get stuck alone in a bot.



Technical Architecture & Security Stack

Pg 2/3

Front End Engineering (Voice & Edge)



AI Core & Cost Efficiency

Translation & AI: IndicTrans 2 paired with Quantized LLM + PostRAG (Answers strictly from Govt Docs).

Semantic Caching: Quadrant / Milvus caches ~70% of repetitive queries. *Result: Reduces compute cost and latency by ~60%.*

3-Layer Auth & Security Gateway

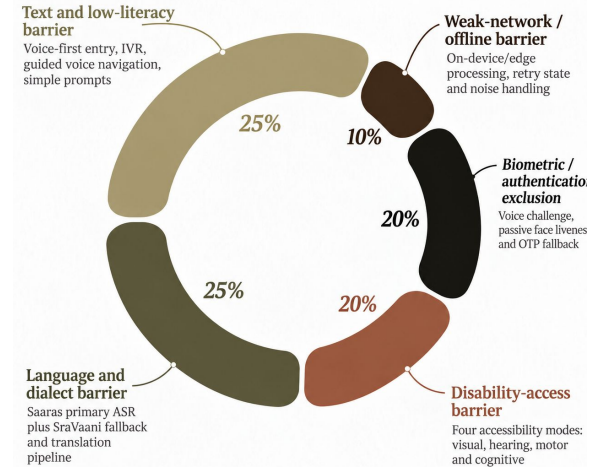
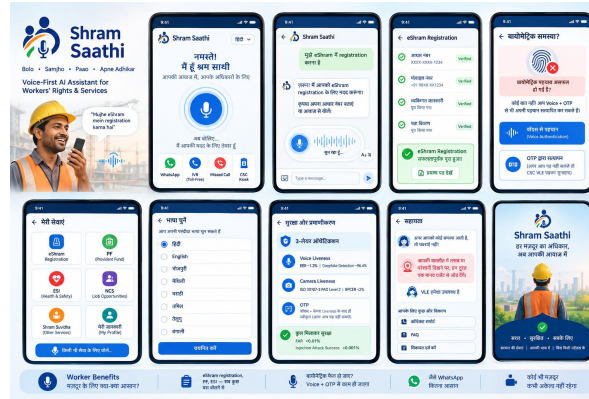
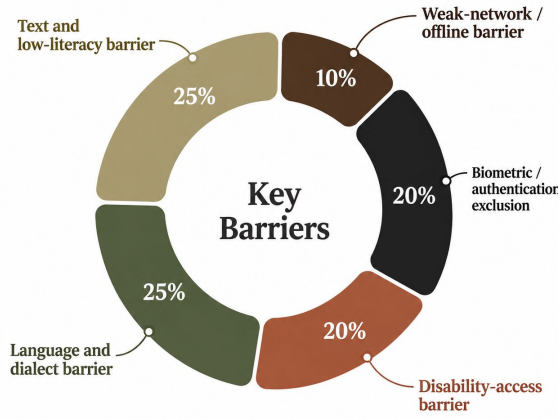
Deep Fake-Proof: Voice Liveness + Camera Liveness + OTP (FAR <0.01%, stops injection attacks).

Data Masking: Aadhaar & sensitive data is auto-masked [XXXX-XXXX] at the API Gateway.

Compliance: TLS 1.3, AES-256-GCM, DPDP 2023 & GIGW 3.0 strictly followed.



APPENDIX



What I have ?

What I create

What I get

Project Links

GitHub Repository: [shram-saathi](https://github.com/shram-saathi) (Source Code & Documentation)

- **Architecture Diagrams:** draw.io & [Mermaid.js](https://mermaid.js)
- **Logic Verification:** Claude Pro (Anthropic)
- **Language Formatting:** ChatGPT (OpenAI)
- **Presentation:** Google Slides

Data Sources & References

Ministry of Labour & Employment (MoLE), Govt of India

- eShram & NCS Official Dashboards
- DARPG BHASHINI Launch Reports (May 2026)